



Tuberculosis

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Tuberculosis is a leading cause of death globally. Given the airborne transmission of tuberculosis, anybody can be infected, but people in high-incidence settings are more exposed. Risk of progression to disease is higher in the first years after infection, and in people with undernourishment, immunosuppression, or who smoke, drink alcohol, or have diabetes. Although cough, fever, and weight loss are hallmark symptoms, people with tuberculosis can be asymptomatic, so a high index of suspicion is required. Prompt diagnosis can be made by sputum examination (ideally with rapid molecular tests), but chest radiography can be helpful. Most people with disease can be treated with regimens of 6 months or less; longer regimens may be necessary for those with drug resistance. Central to successful treatment is comprehensive, person-centred care including addressing key determinants, such as undernourishment, smoking, and alcohol use, and optimising management of comorbidities, such as diabetes and HIV. Care should continue after treatment ends, as long-term sequelae are common. Prevention relies mostly on treatment with rifamycin-based regimens; current vaccines have limited efficacy. Ongoing research on shorter and safer regimens for infection and disease treatment, and simpler and more accurate diagnostic methods will be key for tuberculosis elimination.

Introduction

This Seminar aims to guide health-care providers in responding appropriately and effectively to their patients' concerns, in identifying those individuals who are at risk or who are affected by tuberculosis, and in promptly initiating rapid and accurate investigations and then offering effective preventive or curative treatment with adequate support, including after the successful completion of treatment.

Tuberculosis pathophysiology and risk factors Tuberculosis survivor narrative and perspective

"I was under a lot of stress...trying to figure out how I caught it. How quick can I stop passing it on, 'cause I don't want nobody going through this, what I'm going through."¹

When a person is diagnosed with tuberculosis, one of their first questions is often, "Why did I get tuberculosis? Why me?" A diagnosis of tuberculosis can be overwhelming, and can lead to feelings of shock, shame, and uncertainty. These feelings must be processed while

understanding how this diagnosis will affect one's future and the future of those around them. For some, there is confusion; for others, it might be a relief to have clarity—the cause of the distress might be better understood when accompanied by compassionate and encouraging words of a health-care practitioner and the services to treat the disease. Patients reported that receiving compassionate care from clinicians aided recovery, including an increased sense of responsibility and control over their health.²

Considerations for tuberculosis care providers

The development of tuberculosis disease (panel) can be simplified to a few key steps, with risks of occurrence of each step modified by different factors (figure 1). The possibility of inhaling *Mycobacterium tuberculosis* is determined by the concentration of the bacilli in the air breathed and the duration of exposure.³ Airborne bacterial concentration is higher if the person with tuberculosis has extensive or cavitary pulmonary disease, is sputum-smear positive, or has laryngeal tuberculosis, plus if there are environmental factors that favour accumulation (eg, poorly ventilated indoor environments) over elimination (eg, outdoors or more sunlight).^{3,4} Based on these considerations, household and other close contacts of people with pulmonary tuberculosis are at the highest risk of infection.^{5–7} However, transmission can also occur in the community^{8–11} as many people with pulmonary tuberculosis do not have cough,^{12,13} and *M tuberculosis* can be disseminated simply by typical breathing.¹⁴ The proportion attributable to household versus outside is expected to vary according to the epidemiological situation.

Following inhalation of *M tuberculosis*, infection might be established; 50–70% of people exposed to *M tuberculosis* have negative tuberculin skin tests (TSTs) for tuberculosis infection,^{6,7} this negative reaction and apparent protection from infection might be under genetic control.¹⁵ Although BCG vaccination was long believed to modify risk of

Search strategy and selection criteria

We searched for publications from Jan 1, 2019, to May 1, 2023 initially, and updated June 10, 2024, in the Cochrane database of systematic reviews, and Cochrane registry of randomised trials, using MESH terms for tuberculosis and *Mycobacterium tuberculosis*, as well as TB or tuberc* in title or abstract, combined with MESH terms for diagnosis (diagnosis, bacteriology, microbiology, and radiography), treatment (therapeutics, antitubercular agents, and names of specific tuberculosis drugs), and prevention (antibiotic prophylaxis, vaccination, and contact tracing). We used the reference lists of these papers to identify other relevant randomised trials, large-scale cohort studies, and individual patient data meta-analyses.

Panel: Definitions of common tuberculosis-related terms

Tuberculosis disease

Presence of macroscopic pathology caused by *Mycobacterium tuberculosis* complex organisms, including *M tuberculosis* (the majority), *Mycobacterium bovis*, *Mycobacterium orygis*, *Mycobacterium africanum*, or *Mycobacterium caprae*. People might or might not be contagious and might or might not have symptoms. Treatment is given to prevent morbidity and mortality, to reduce transmissibility, and to improve symptoms when present.

Tuberculosis infection

Also referred to as latent tuberculosis infection, or *M tuberculosis* infection. Presence of viable *M tuberculosis* and a host response. People with tuberculosis infection neither have symptoms or physical signs, nor microbiological or macroscopic pathological results consistent with tuberculosis disease. These people are not contagious but are at risk of developing tuberculosis disease.

Tuberculosis preventive treatment

Treatment given to a person with presumed tuberculosis infection to lower the risk of future tuberculosis disease.

Drug-susceptible tuberculosis

Disease due to strains susceptible to all first-line tuberculosis drugs (eg, isoniazid, rifampicin, ethambutol, and pyrazinamide).

Drug-resistant tuberculosis

Disease due to *M tuberculosis* resistant to at least one of the four first-line tuberculosis drugs. Strains resistant to isoniazid but susceptible to rifampicin are isoniazid-monoresistant tuberculosis, those resistant to rifampicin but susceptible to isoniazid are rifampicin-monoresistant tuberculosis, and those resistant to both isoniazid and rifampicin are multidrug-resistant tuberculosis. Multidrug-resistant strains also resistant to a fluoroquinolone are defined as pre-extensively drug-resistant tuberculosis, while extensively drug-resistant tuberculosis is defined as resistance to rifampicin and isoniazid plus fluoroquinolones and either linezolid or bedaquiline.

Extrapulmonary tuberculosis

Although it occurs most frequently in the lungs (pulmonary tuberculosis), tuberculosis disease can occur in any other organ including skin, bones, lymph nodes, pleura, heart, gastrointestinal tract, and CNS. These forms are rarely contagious, but cause considerable difficulty and mortality, especially in children who are also more susceptible to disseminated disease.

developing disease without preventing infection, several studies have shown that BCG-vaccinated contacts are less likely to have positive interferon- γ release assays (IGRAs),^{16–18} implying some protection from infection.

The risk of progression from tuberculosis infection to disease is affected by time since infection and host susceptibility (figure 2). The highest risk of disease is in the first year after infection, and decreases thereafter,^{4,19,20} as seen in household contacts and in recent migrants from high-incidence countries to low-incidence countries.^{21,22} Among contacts, the risk is highest in young children, and lower in adolescents and adults.^{22,23} Beyond time and age, risk factors for progression to tuberculosis disease can be conceptualised at an individual level or a public health level. For an individual with a positive TST or IGRA, risk of disease is increased if they have HIV infection, silicosis, end-stage kidney disease requiring dialysis, or are taking immunosuppressive medications (table 1). On the other hand, even though the individual risk associated with undernutrition, smoking, alcohol consumption, and diabetes is not as high, these four conditions have a major effect on public health globally, because they are so common.³⁸

Epidemiology

Tuberculosis was the leading infectious cause of death worldwide in 2023, with 1·3 million estimated deaths, surpassing both COVID-19 and HIV/AIDS.³⁸ An estimated 10·6 million people developed tuberculosis,

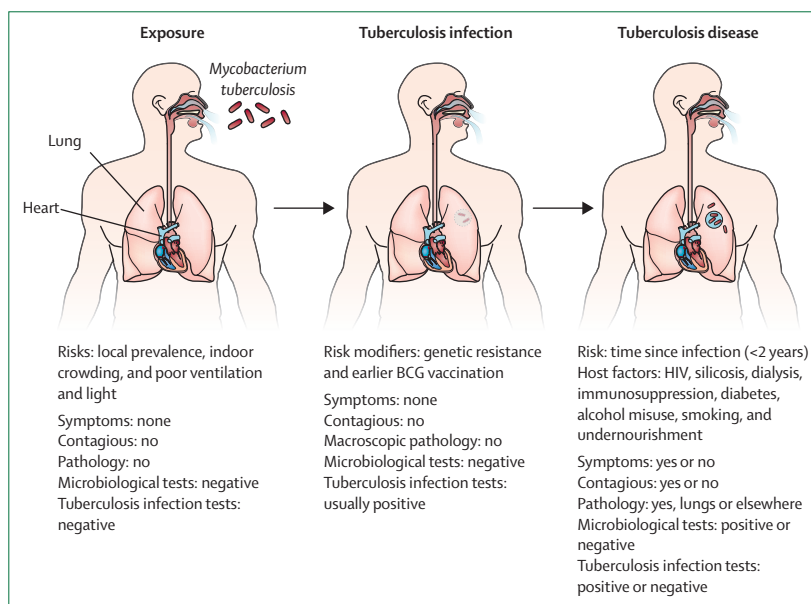


Figure 1: Risk factors and main clinical characteristics of tuberculosis infection and disease

Microbiological tests include smear microscopy, molecular tests, or culture. Tuberculosis infection tests include tuberculin skin tests, tuberculosis-specific skin tests, and interferon- γ release assay.

although only 7·5 million were reported to WHO;³⁸ this gap is largely due to undiagnosed tuberculosis. In total, 1·3 million children were estimated to have tuberculosis disease, of whom more than half (51–58%) were missed.³⁸

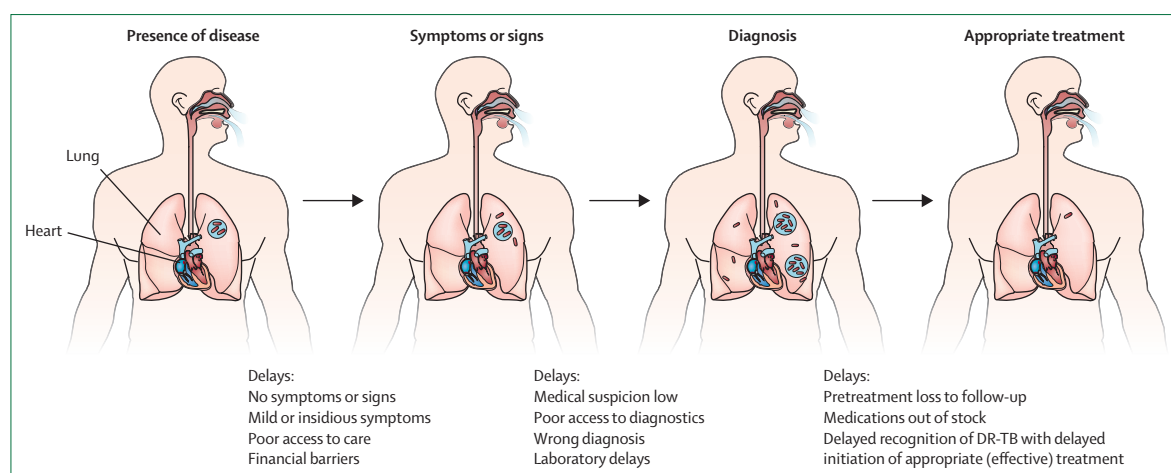


Figure 2: Tuberculosis cascade of care
DR-TB=drug-resistant tuberculosis.

The detection gap also includes over 275 000 people with drug-resistant tuberculosis. People living with HIV are at particular risk for tuberculosis disease, and accounted for 6·3% of all incidence of tuberculosis and 13% of tuberculosis-related deaths in 2022.³⁸ Estimated rates of tuberculosis disease vary widely in different countries, with the highest rates and greatest burden of disease in low-income and middle-income countries. Rates are highest in countries in Africa and southeast Asia, followed by the Western Pacific.³⁸ In high-income countries, most people who develop tuberculosis disease are migrants; rates in these populations parallel rates in their countries of origin.³⁹

The effect of COVID-19 on the epidemiology and clinical aspects of tuberculosis must also be acknowledged. The response to COVID-19 might have reduced tuberculosis transmission due to less person-to-person contact, but also worsened outcomes due to delayed diagnosis and poor access to tuberculosis care.⁴⁰ There is great uncertainty regarding the overall effect, which could vary within and between countries.⁴¹

Clinical presentation, signs and symptoms, and diagnostic investigations

Tuberculosis survivor narrative and perspective

“At first, I thought I was the only person who had spent months being misdiagnosed—however, when I attended my first TB survivor workshop, I learned the unfortunate truth that my experience was not unique. I heard story after story of misdiagnosis and delayed diagnosis.”

Tenzin Kunor (WHO Global Tuberculosis Report Press Conference; Online, Oct 14, 2021).

“I was very happy because my disease was found before my death.”⁴²

Delayed or incorrect diagnoses often means unnecessary difficulties for the person with tuberculosis. It also means others, such as household and close

contacts, are vulnerable to tuberculosis. Everyone is deserving of accessing diagnostic services and receiving treatment for this curable and preventable disease. One crucial issue is that health-care providers often do not consider tuberculosis when working with patients and, despite best practices established around when to test for tuberculosis, opportunities are too often missed. One key role a practitioner can take is helping the person with tuberculosis better understand what a diagnosis of tuberculosis means, to help make sense of this experience.

Considerations for tuberculosis care providers

Tuberculosis is usually detected in people who seek medical attention because they have symptoms (figure 3). In people with lung involvement, the most common symptom is cough, although systemic symptoms such as fever or weight loss are common. Among children, the only symptoms of tuberculosis might be poor weight gain and lethargy.⁴³ Symptoms of extrapulmonary tuberculosis reflect the site involved, which can be any organ.^{44,45} It is important to be aware that people with pulmonary tuberculosis might be asymptomatic,^{8,12,13} and so tuberculosis can be detected on a chest radiograph performed for other reasons; extrapulmonary disease can also be asymptomatic^{44,45} (eg, uterine tuberculosis detected as part of an infertility investigation). Children are more prone to extrapulmonary tuberculosis, including disseminated tuberculosis and CNS tuberculosis, which are medical emergencies and need rapid recognition and appropriate referral.⁴⁶ It is often stated that assessment for tuberculosis should be initiated only if symptoms have been present for over 2 weeks. However, in people at increased risk of tuberculosis disease (due to recent infection or greater susceptibility), symptoms of any type or duration should prompt health providers to think of tuberculosis.

Time from onset of symptoms to tuberculosis diagnosis ranges from 1 month to 12 months in different settings.⁴⁷

This range reflects the insidious nature of symptoms, plus scarce access to care, so many people defer seeking care (patient delay). Providers often do not think of tuberculosis and so are slow to order appropriate tests (provider delay), and receiving results of these tests could take additional weeks (laboratory delay).⁴⁷ Long delays in diagnosis are common for extrapulmonary tuberculosis, for different reasons: in low-incidence settings, these forms are rare and could mimic other diseases; and in high-incidence, low-resource countries, imaging, biopsies, and histopathological examination might not be accessible.

Chest radiography is often the first test ordered for investigation of respiratory symptoms, and although specificity is only 54–60%,⁴⁸ this test can help identify the presence of pulmonary disease. One restriction is the need for a specialist to interpret radiographs; this limitation can be overcome with use of computer-aided detection (CAD) algorithms for automatic reading.⁴⁹ In people with symptoms, most CAD products have accuracy equivalent to that of experienced radiologists,^{48,50} but are less accurate in people who are asymptomatic, older, living with HIV, have a history of previous tuberculosis, or have smear-negative tuberculosis disease.⁵⁰ Another restriction is the absence of validated CAD products for children.

In other settings, sputum for microscopy, culture, or molecular testing is the first test. Smear microscopy for acid-fast bacilli is still widely used as it is simple, rapid (results within 1 day), identifies people with more contagious tuberculosis, and can be used to monitor response to treatment. Key limitations are low sensitivity of the test to detect pulmonary tuberculosis disease and the need for more information on drug resistance. Culture in liquid media, followed by phenotypic drug susceptibility testing has long been the reference standard for diagnosis of tuberculosis, but requires complex laboratory infrastructure, and results can take weeks to months for detection and susceptibility testing,^{51,52} especially in people with less advanced disease.⁵¹ In addition, drug susceptibility testing is difficult or unreliable for some drugs, notably pyrazinamide.⁵³

WHO now recommends that the first test for patients with presumptive pulmonary tuberculosis is a rapid molecular diagnostic test that amplifies *M tuberculosis* nucleic acid.⁵⁴ These tests (table 2) provide results within a day, such as microscopy, but have much higher sensitivity, approaching that of culture in pulmonary tuberculosis,⁵⁸ and are also useful for diagnosis of extrapulmonary forms.⁵⁹ Their use has been associated with improved treatment outcomes, including lower mortality from tuberculosis.^{60,61} As a result, these tests are fast replacing acid-fast bacilli smear in many settings, including low-income and middle-income countries.³⁸

Young children often do not provide sputum for examination and have paucibacillary disease, making

	Annual risk of tuberculosis for the first 2–3 years after a positive TST or IGRA*	Reference
Very high risk		
People with HIV†	1.7–2.7%	Gupta et al ²¹ Campbell et al ²⁴
Child or adolescent (<18 years) tuberculosis contact	2.9–14.6%	Gupta et al ²¹ Martinez et al ²³
Adult (≥18 years) tuberculosis contact	0.8%–3.7%	Gupta et al ²¹ Campbell et al ²⁴
Silicosis	3.7%	Campbell et al ²⁴
High risk		
Stage 4 or 5 chronic kidney disease with or without dialysis	0.3–1.2%	Campbell et al ²⁴
Transplant recipients (solid organ or haematopoietic)	0.1–0.7%	Campbell et al ²⁴
Fibronodular disease	0.2–0.6%	From incidence rates in three longitudinal studies; Grzybowski et al ^{25,26} Nolan and Elarth ²⁷
Receiving immunosuppressing drugs (eg, tumour necrosis factor α inhibitors or steroids)‡	0.5%	Campbell et al ²⁴
Cancer (lung cancer, sarcoma, leukaemia, lymphoma, or gastrointestinal cancer)	0.1–0.4%	Estimated from hazard ratio in a population-based study; Kumar et al ²⁸
Moderate risk		
Granuloma on chest x-ray	0.1%	From incidence rates in two longitudinal studies; Grzybowski et al ²⁵ Horwitz and colleagues ²⁹
Diabetes	0.1–0.2%	Estimated from pooled relative risk in observational studies; Jeon and Murray ³⁰
Undernutrition	0.1%	Estimated from pooled relative risk in observational studies; Franco et al ³¹
Heavy alcohol use (≥3 drinks per day)	0.1–0.2%	Estimated from pooled relative risk in observational studies; Lönnroth et al ³²
Heavy tobacco cigarette smoker (≥1 pack per day)	0.1%	Estimated from pooled odds ratios and relative risks in two meta-analyses of observational studies; Lin et al ³³ Bates et al ³⁴
Low risk		
General, adult population with no known risk factor	0.03%	Campbell et al ²⁴
People with a positive two-step TST booster and no known risk factor	0.02%	Extrapolated from a longitudinal study and a randomised trial; Comstock et al ³⁵ Ferebee SH ³⁶

TST=tuberculin skin test. IGRA=interferon-γ release assay. *Risks are presented as point estimates for ease of presentation; ranges, when presented, are derived from multiple studies or different tests used (TST or IGRA). Where possible, absolute risks are derived from large systematic reviews; where absolute risks were not available, but relative risks were, we multiplied the general population absolute risk by the relative risk to approximate absolute risk. Risks are expected to diminish after the first 2–3 years, although in settings where frequency of exposure is high, risks might continue to be elevated.²¹ †Studies informing this estimate are from the antiretroviral era; proportion receiving antiretroviral therapy in these studies ranged from 40% to 75%. ‡Risk does not appear substantially elevated with low-dose steroids (ie, prednisone), but elevated with moderate or high dose (low dose ≤9 mg/day; medium dose 10–19 mg/day; and high dose ≥20 mg/day).³⁷

Table 1: Risk of tuberculosis disease among different populations with a positive test for tuberculosis infection

microbiological confirmation challenging. Hence diagnosis of tuberculosis disease in children is best

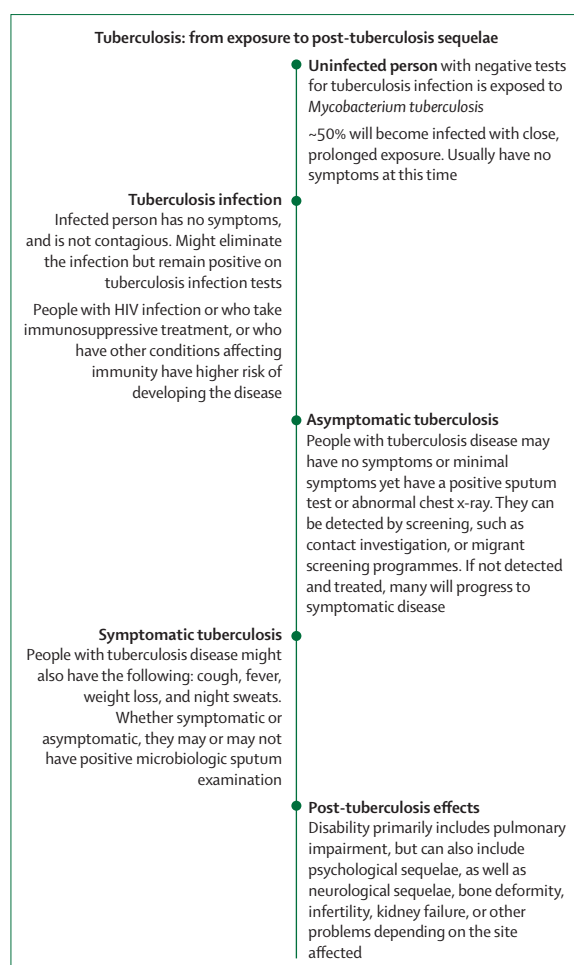


Figure 3: Progression of tuberculosis infection from initial exposure to post-tuberculosis sequelae

made using a combination of signs, symptoms, chest radiography findings, and integrated treatment decision algorithms.⁶² Induced sputum, gastric lavage, nasopharyngeal aspirate, or stool samples should be tested first with molecular tests and then with culture (although not stool samples) if molecular tests are negative or indicate drug resistance.^{54,62}

A decision to test entails a responsibility to provide treatment, should the test be positive. Despite this fundamental principle, approximately 15% of people with a test confirming tuberculosis disease are not started on treatment.⁶³ Optimisation of health-service delivery to assure adequate follow-up is an obvious solution, but there is also increasing global interest in the development and deployment of affordable point-of-care tests.⁶⁴ Tests that do not need sample transportation to a central laboratory, use simple analytic processes, and provide results very rapidly⁶⁵ could close the gap between diagnosis and treatment. At present, the lateral flow lipoarabinomannan AlereLAM urine test is the only point-of-care test recommended by WHO, but use is

restricted to people living with HIV who have CD4-positive cell counts under 100 per μL , have CD4-positive cell counts under 200 per μL plus symptoms, or are hospitalised.^{66,67}

Acute management: treatment, support, and follow-up of people with tuberculosis disease Tuberculosis survivor narrative and perspective

"The medication for TB should not cause us harm. I put up with my medication because I want to be cured. I wish someone would have just explained to me what the problem was and the drugs that they were giving me."⁶⁸

Treatment for tuberculosis disease has an immense effect on people's lives in a multitude of ways. It can affect one's physical health with various possible side-effects, such as chronic fatigue, nausea, somatic sensations, or even irreversible disabilities. Providers should be equally mindful of how treatment can affect a person's psychological health. Physical and role functioning, experience with social stigmatisation, isolation, pill burden, and treatment duration are all examples. Shorter treatment options have the possibility to mitigate some of the burden on people with tuberculosis disease, but safety and acceptability of these shorter regimens cannot be forgotten. Quality care in treating people with tuberculosis disease needs more than a biomedical approach. Person-centred care includes integrated psychosocial support care, peer support, and referrals within and beyond the health sector. People might have numerous questions and concerns, which require deep listening, responsiveness, and constant evaluation of regimen appropriateness.

Considerations for the tuberculosis care providers

People with tuberculosis, including those diagnosed during pregnancy,^{69,70} should be treated promptly and followed up monthly for adherence, side-effects, and general support.⁷¹ The standard treatment regimen for drug-susceptible tuberculosis is 6 months of rifampicin and isoniazid, supplemented with pyrazinamide and ethambutol in the first 2 months (2HRZE-4HR).⁷¹ Drugs should be taken daily throughout treatment—although taking doses three times per week during the last 4 months is acceptable,⁷² and can be taken with pyridoxine to reduce the risk of isoniazid-induced peripheral neuropathy. Although the standard regimen is efficacious, treatment-related grade 3–4 adverse events occur in approximately 5–10% of participants in clinical trials.^{73–75} The most important and potentially life-threatening adverse event is hepatotoxicity, which can be due to pyrazinamide (highest risk), isoniazid, or rifampicin.⁷⁶

People who are pregnant or breastfeeding have been excluded from almost all trials of treatment of tuberculosis disease and tuberculosis infection. Hence, there is scant evidence on which to base

	Sensitivity for tuberculosis diagnosis (sputum, except LAM or CAD)*	Specificity for tuberculosis diagnosis (sputum, except LAM or CAD)*	Drug resistance detected	Sensitivity for drug resistance	Specificity for drug resistance
For initial evaluation in adults, children, PLHIV without a history of previous tuberculosis, or with remote history of tuberculosis (>5 years) testing of sputum samples (except LAM and x-ray)					
Sputum smear microscopy†	0.60 to 0.90 ^{55,56}	0.90 to 0.98	NA	NA	NA
Chest X-ray with or without CAD‡§	0.71 to 0.93	0.64 to 0.80	NA	NA	NA
Xpert MTB/Rif	Overall: 0.85 (0.99 in smear-positive samples and 0.61 in smear-negative samples)	0.98	Rifampicin	0.95	0.99
Xpert Ultra ⁵⁷	Overall: 0.91 (0.99 in smear-positive samples and 0.78 in smear-negative samples)	0.96	Rifampicin	0.95	0.99
Truenat MTB-Rif Dx	Overall: 0.80 (0.96 in smear-positive samples and 0.46 in smear-negative samples)	0.96	For those in whom Truenat was positive for tuberculosis (two-step test), rifampicin	0.84	0.97
Abbott RealTime TB	0.93§	0.98§	Rifampicin	Rifampicin: 0.97§	Rifampicin: 0.99§
Abbott RealTime TB RIF/INH	0.93§	0.98§	Rifampicin and isoniazid	Rifampicin: 0.97§; isoniazid: 0.86§	Rifampicin: 0.99§; isoniazid: 0.99§
BD MAX MDR-TB	0.93§	0.98§	Rifampicin and isoniazid	Rifampicin: 0.97§; isoniazid: 0.86§	Rifampicin: 0.99§; isoniazid: 0.99§
Cobas MTB	0.93§	0.98§	Rifampicin	Rifampicin: 0.97§	Rifampicin: 0.99§
Cobas MTB RIF/INH	0.93§	0.98§	Rifampicin and isoniazid	Rifampicin: 0.97§; isoniazid: 0.86§	Rifampicin: 0.99§; isoniazid: 0.99§
FluoroType MTB	0.93§	0.98§	-	-	-
FluoroType MTBDR	0.93§	0.98§	Rifampicin and isoniazid	Rifampicin: 0.97§; isoniazid: 0.86§	Rifampicin: 0.99§; isoniazid: 0.99§
TB-LAMP	0.78	0.98	-	-	-
For initial evaluation of PLHIV of all ages with symptoms and CD4 under 200 per µL or CD4 over 100 per µL without symptoms‡¶					
AlereLAM	0.54¶	0.88¶	-	-	-
For resistance detection if initial rifampicin resistance, or exposed to isoniazid-resistant, MD-R or XDR-R tuberculosis, presumptive treatment failure, or history of tuberculosis within 5 years					
Culture with phenotypic drug susceptibility tests	Considered the reference (assumed 100%)	..	All	Considered the reference (assumed 100%)	
GenoType MTBDRplus v1 (Hain 1)	Not recommended in smear-negative samples	..	Rifampicin and isoniazid	Rifampicin: 0.97; isoniazid: 0.88	Rifampicin: 0.98; isoniazid: 0.98
GenoType MTBDRplus v2 (Hain 2)	Not recommended in smear-negative samples	..	Rifampicin and isoniazid	Rifampicin: 0.96; isoniazid: 0.95	Rifampicin: 0.98; isoniazid: 0.99
GenoType MTBDRsl for second-line drugs (version 2 available, new data awaited)	Not recommended in smear-negative samples	..	Fluoroquinolones, second-line injectables, and XDR	Fluoroquinolones: 0.86; second-line injectables: 0.77; XDR: 0.71	Fluoroquinolones: 0.99; second-line injectables: 0.99; XDR: 0.99
Genoscholar NTM+MDRTB II (Nipro)	Not recommended in smear-negative samples	..	Rifampicin and isoniazid	Rifampicin: 0.75 to under 0.99; isoniazid: 0.50 to 0.95	Rifampicin: 0.97 to under 0.99; isoniazid: 0.97 to 0.98
Genoscholar PZA-TB II	Not recommended in smear-negative samples	..	Pyrazinamide	0.81	0.98
Xpert XDR	Not recommended for diagnosis	..	Isoniazid, ethionamide, fluoroquinolones, amikacin, capreomycin, and kanamycin	Isoniazid: 0.94; ethionamide: 0.98; fluoroquinolones: 0.93; amikacin, capreomycin, and kanamycin: 0.94	Isoniazid: 0.98; ethionamide: under 0.99; fluoroquinolone: 0.98; amikacin, capreomycin, and kanamycin: 0.98

LAM=Lipoarabinomannan. CAD=computer-aided detection for interpretation. PLHIV=people living with HIV. NA=not applicable. MDR=multidrug resistant. XDR=extensively drug resistant (the older definition is used here, meaning MDR plus resistance to a fluoroquinolone and a second-line injectable). *In adults with presumptive tuberculosis, compared with culture with phenotypic drug susceptibility tests. For other specific populations (PLHIV or children) or samples (extrapulmonary samples or culture isolates), please refer to reference.⁵⁴ For predictive value in settings or populations with different prevalence, also refer to reference.⁵⁴ In young children that do not provide sputum, tests should be performed in gastric aspirate, nasopharyngeal aspirate, or stool; †WHO no longer recommends smear microscopy as the initial test, but smear microscopy is still used in many settings, where it might be the only test available; ‡Use of different CAD interpretation thresholds can increase sensitivity but reduce specificity, or vice versa (see text). For use in different clinical situations and populations, and for different software or versions, see reference.⁵⁴ Not evaluated in children. Accuracy is lower in PLHIV and previously treated people; §Accuracy for all tests pooled in the systematic review;⁵⁴ ¶|| In PLHIV with symptoms and CD4 cell count under 100 cells per µL. Sensitivity is lower if CD4 count is over 100 cells per µL, and much lower if CD4 is over 200 per µL. Sensitivity is also lower if patient is asymptomatic (see reference⁵⁴ for more information); || Data from tests in isolates, not enough evidence from data in sputum.

Table 2: Summary of tests for tuberculosis disease diagnosis and drug-resistance detection (accuracy of tests taken from systematic reviews completed for WHO guidelines⁵⁴)

recommendations, although observational studies and programmatic experience have led to acceptance that standard therapy can be given.⁷¹ A 2019 trial of isoniazid preventive treatment among women with HIV who are pregnant or postpartum revealed high rates of severe toxicity including deaths,⁷⁷ reinforcing the need to include this group in future trials.⁷⁸

The past 20 years have seen tremendous efforts to shorten drug-susceptible tuberculosis treatment, although given the substantial toxicity of the standard regimen, a regimen that is safer—and not simply shorter—is needed.⁷⁹ Three trials consistently showed that 4-month fluoroquinolone-based regimens had inferior efficacy when compared with the standard 6-month regimen.^{73,75,80} SimpliciTB, a trial examining a 4-month regimen of bedaquiline, pretomanid, pyrazinamide, and moxifloxacin, also did not show non-inferior efficacy, and raised significant safety concerns.⁸¹ In a trial of 2343 participants, a 4-month regimen of isoniazid, rifapentine, and moxifloxacin, with pyrazinamide for the first 2 months (2HPMZ–2HPM) had non-inferior effectiveness compared with the standard 6-month regimen, but more frequent treatment-related grade 3–4 adverse events (12·9% vs 9·8%, respectively).⁷⁴ In two subsequent small studies in San Francisco (CA, USA) and New York (NY, USA), 11 (50%) of 22, and 23 (66%) of 35 people did not complete this 4-month regimen, largely because of poor tolerability and safety.^{82,83} In the TRUNCATE TB trial, participants received only 2 months of treatment with bedaquiline, linezolid, isoniazid, pyrazinamide, and ethambutol and then were followed up closely for relapse.⁸⁴ Recurrence rates were unacceptably high, and toxicity remains a concern, but the trial did show the potential of new tuberculosis drugs for treatment shortening. Other efforts have focused on increasing the dose of rifampicin, as the dose currently recommended (10 mg/kg of bodyweight) might be suboptimal.^{85,86} In a recent trial (RIFASHORT), double and triple doses of rifampicin for 4 months did not show non-inferior efficacy;⁸⁷ other studies are ongoing. Based on the SHINE-TB trial⁸⁸ it is reasonable to give 4 months of a regimen with isoniazid, rifampin, pyrazinamide, and ethambutol, all at standard doses, to children with less severe tuberculosis (smear-negative).

Therapeutic drug monitoring has been used to optimise drug concentrations,^{89,90} but the clinical benefits of this approach are not yet confirmed. Risk stratification has been suggested to optimise duration; in post-hoc analyses of three trials,^{73,75,80} clinical characteristics were used to classify people into three groups: some could be cured with 4 months, others with 6 months, whereas some required more than 6 months to achieve relapse-free cure.⁹¹ The SHINE-TB trial found that a 4-month regimen (2HRZE–2HR) had non-inferior efficacy compared with the standard 6-month regimen in children aged

3 months to 16 years with non-severe (smear-negative) drug-susceptible tuberculosis.⁸⁸ This regimen is now recommended by WHO.⁷¹ In children and adolescents with extrapulmonary tuberculosis, the standard regimen is recommended, although a 6-month intensive regimen (6HRZEto) could be used for those with tuberculosis meningitis.⁶²

Treatment of drug-resistant forms of tuberculosis remains a major challenge. In 2022, only 63% of people with disease due to rifampicin-resistant tuberculosis had a successful treatment outcome compared with 88% of people with drug-susceptible tuberculosis.³⁸ Isoniazid monoresistance is the most common form of drug-resistant *M tuberculosis* but testing for resistance to isoniazid is not available in many settings. Treatment of unrecognised isoniazid monoresistance with the standard 6-month regimen is associated with statistically significant higher rates of failure, disease recurrence, and acquired drug resistance, especially multidrug resistance.⁹² WHO recommends a 6-month regimen of rifampicin, pyrazinamide, and ethambutol, plus a fluoroquinolone, based on an individual participant data meta-analysis of observational studies.⁹³

The treatment of multidrug-resistant tuberculosis or rifampicin-resistant tuberculosis has been revolutionised with the development of the 6-month all-oral bedaquiline, pretomanid, linezolid, with or without moxifloxacin (BPAL or BPALM, respectively) regimens. In the TB-PRACTECAL trial, BPALM was statistically significantly safer, and resulted in a statistically significant higher rate of favourable treatment outcomes than the standard WHO-recommended regimens.⁹⁴ Importantly, BPAL appeared as efficacious as BPALM, making it a good option for fluoroquinolone-resistant tuberculosis. The 6-month BPAL regimen had similar efficacy in the NIX⁹⁵ and ZENIX⁹⁶ trials, but toxicity was common, particularly peripheral neuropathy, with linezolid doses of 1200 mg/day. From these three studies, it appears that BPAL or BPALM regimens should now be considered first-line treatment for rifampicin-resistant tuberculosis,⁹⁷ but the dose and duration of linezolid to achieve the optimal balance of safety and efficacy is uncertain. The inclusion of patients as young as 14 years in these trials allowed WHO to extend the recommendation to this age group;⁹⁷ there is no high-quality evidence on use of these regimens in younger children and pregnant people.⁹⁸ The pharmacokinetics and safety of pretomanid in children will be evaluated in the IMPAACT 2034 trial (NCT05586230). Other short, safe, effective options for rifampicin-resistant tuberculosis are being tested in ongoing trials.

For people with multidrug-resistant tuberculosis or rifampicin-resistant tuberculosis with pretomanid or linezolid intolerance or resistance, additional regimens include a 9–11-month all-oral regimen with bedaquiline, which has been evaluated in programmatic settings.⁹⁹ Regimens of 18–20 months, composed of multiple group

Considerations	
Treatment regimen(s)	
Tuberculosis infection	
6–9 months of daily isoniazid	Completion poor due to longer duration. Important and potentially fatal hepatotoxicity. Should be considered second-line treatment.
3 months of once weekly isoniazid and rifapentine	Can be delivered directly observed or self-administered; randomised trial evidence that completion better with directly observed administration. Not for children aged under 2 years. Important toxicity of systemic immune reaction syndrome.
3 months of daily isoniazid and rifampicin	Formulations used are the same as used for tuberculosis disease treatment. This formulation enhances access and feasibility. However, toxicity of isoniazid is combined with drug–drug interactions of rifampicin, and it is no safer than 6–9 months of isoniazid.
4 months of daily rifampicin	Safest of all currently recommended regimens, with rash the most common adverse event. Worse completion than the other rifamycin-containing regimens.
1 month of daily rifapentine and isoniazid	Only recommended for PLHIV as randomised trial only conducted in this population; trials among people without HIV are ongoing.
36 months of daily isoniazid	Only recommended for PLHIV in settings of high tuberculosis transmission. Poor completion and high rates of toxicity.
6 months of daily levofloxacin	For the treatment of contacts of or rifampicin-resistant tuberculosis or multidrug-resistant tuberculosis index patients. Two placebo-controlled trials have assessed levofloxacin in adults and children (ACTRN12616000215426 and ISRCTN92634082). Good safety profile. Efficacy similar to 6 months of isoniazid.
Drug-susceptible pulmonary tuberculosis disease	
2 months of daily isoniazid, rifampicin, pyrazinamide, and ethambutol, followed up by 4 months of daily isoniazid and rifampicin	Standard regimen worldwide. Recommended for all ages and for people who are pregnant (although in some countries pyrazinamide is not recommended for people who are pregnant, or for people older than 70). Considerable toxicity (5–10% of adults will have to stop at least one drug for a severe adverse event), and 3–4% relapse rates.
2 months of daily isoniazid, rifampicin, pyrazinamide, and ethambutol, followed up by 2 months of daily isoniazid and rifampicin	Only recommended for children aged 3 months to 16 years with non-severe smear-negative, probable drug-susceptible tuberculosis. Non-severe defined as respiratory tuberculosis confined to one lobe (opacification of <1 lobe) with no cavities, no signs of miliary tuberculosis, no complex pleural effusion, and no clinically significant airway obstruction or peripheral lymph-node tuberculosis.
2 months of daily isoniazid, rifapentine, moxifloxacin, and pyrazinamide, followed up by 2 months of daily isoniazid, rifapentine, and moxifloxacin	Only recommended for people aged 12 years or over, based on a single non-inferiority trial in adolescents and young adults. Tolerability might be worse than with standard regimen.
Isoniazid-resistant pulmonary tuberculosis disease	
6 months of rifampicin, pyrazinamide, ethambutol, and levofloxacin	Only evidence base is an individual participant data meta-analysis of observational data. Efficacy similar with or without the inclusion of isoniazid; safety data are scarce. One trial to assess regimen with levofloxacin vs without levofloxacin ongoing.
Rifampicin and multidrug-resistant pulmonary tuberculosis disease	
6 months of bedaquiline, pretomanid, linezolid, and moxifloxacin	Preferred if fluoroquinolone susceptible. Bedaquiline given daily for 2 weeks, before being switched to three times weekly; all other drugs given daily. Linezolid should be given at a dose of 600 mg but can be reduced to 300 mg or discontinued if toxicity emerges. Approved for children aged 14 years and older. No data for children younger than 14 years or people who are pregnant. Key toxicities with component drugs: bedaquiline, QT prolongation and hepatotoxicity; pretomanid, hepatotoxicity; linezolid, peripheral neuropathy and neutropenia; moxifloxacin, gastrointestinal toxicity and QT prolongation. Shorter and safer than previous standard of care regimens in one trial, but treatment-related grade 3–4 adverse events in up to 20% of people.
6 months of bedaquiline, pretomanid, and linezolid (BPaL)	Preferred if patient is resistant to fluoroquinolone. Dosing and safety information as for bedaquiline, pretomanid, linezolid, and moxifloxacin. Approved for children aged 14 years and older. No data for younger children or people who are pregnant.
9 months of daily levofloxacin or moxifloxacin, clofazimine, ethambutol, and pyrazinamide, supplemented with high-dose isoniazid for the first 4–6 months, bedaquiline in the first 6 months, and either linezolid for the first 2 months or ethionamide for the first 4–6 months	Preferred when pretomanid or linezolid are contraindicated. Bedaquiline given daily for 2 weeks, before being switched to thrice weekly; all other drugs given daily. Linezolid should be given at a dose of 600 mg but can be reduced to 300 mg or discontinued if toxicity emerges. More effective than injectable-containing regimens of similar duration; safety is expected to be better but has not been evaluated.
18–20 months of a 4–5 drug regimen comprising, where possible, all three Group A agents and at least one Group B agent; if only two Group A agents can be used, at least two Group B agents should be included; if a four-drug regimen of probably effective drugs cannot be comprised with Group A and B agents, Group C agents should be used.	Use if resistance or contraindications to bedaquiline or to fluoroquinolone and linezolid, and other short regimens cannot be used. The medications and duration of this regimen make it more toxic than shorter regimens. Group A: moxifloxacin or levofloxacin, linezolid, and bedaquiline. Group B: cycloserine or terizidone and clofazimine. Group C: ethambutol, delamanid, pyrazinamide, imipenem–cilastatin or meropenem, amikacin or streptomycin, ethionamide or prothionamide, and p-aminosalicylic acid. Avoid use of any drug for which resistance is detected on drug susceptibility testing.

Table 3: Treatment regimens for tuberculosis infection and disease

B and so-called group C drugs, are needed to treat extensively drug-resistant tuberculosis¹⁰⁰ (table 3). Bedaquiline and delamanid can be used in children of all ages as part of longer regimens.⁶²

Person-centred care during tuberculosis treatment

For many years, directly observed therapy (DOT) was recommended by WHO to ensure adherence. However,

systematic reviews have concluded that end-of-treatment outcomes are not different with DOT than with self-administered therapy,^{101–103} with no difference between community-based DOT and clinic-based DOT.¹⁰⁴ In addition to scarce evidence of effectiveness, mandatory DOT does not align with principles of patient-centred care.¹⁰⁵ More flexible methods to support adherence have been introduced, such as the use

of digital health technologies,¹⁰⁶ although their effectiveness has been variable.¹⁰⁶ Other interventions that could improve tuberculosis treatment outcomes plus have broader benefits include enablers, incentives, and other forms of patient support,¹⁰⁷ addressing psychosocial sequelae,¹⁰⁸ conditional cash transfer schemes,¹⁰⁹ food supplements,¹¹⁰ and other social-protection interventions.^{111,112} Providers should assess and intervene on known determinants of tuberculosis, including undernutrition (especially given the prognostic significance¹¹³), smoking, alcohol, and diabetes.¹¹⁴ In settings with high tuberculosis burden, WHO recommends family-centred, integrated models of care to deliver diagnostic and treatment services close to the patient's home for children and adolescents with signs and symptoms of, or exposure to, tuberculosis.¹¹⁵

Because tuberculosis is highly stigmatised and has major financial consequences, an important role of care providers is education of people with tuberculosis, their families, and communities to reduce stigma,¹¹⁶ and organisation of social and financial support. This approach is best provided through coordinated care from a multidisciplinary care team.¹¹⁷ The common practice of isolation (although no longer recommended), the long duration of treatment, and the disability due to disease will together result in missed work or school, with one in four people with tuberculosis losing their jobs altogether.^{97,118,119} Together, these factors result in substantial direct (out-of-pocket) and even more substantial indirect costs from lost income.^{97,120} For almost half of all people with tuberculosis, these costs will total more than 20% of their annual income (defined as catastrophic costs),¹¹⁹ and are very important barriers to care engagement. Patient costs are higher for people living with HIV, people with drug-resistant forms of tuberculosis, or when the patient with tuberculosis is the head of the family, but lower when tuberculosis is diagnosed through active case finding such as contact investigation.^{97,119}

Tuberculosis prevention

Tuberculosis survivor narrative and perspective

"The very first stigma I experienced was from a close circle of friends and family members whose reactions were just pure shock. They immediately suggested that I hide my diagnosis... This made me feel really guilty, ashamed of myself for getting tuberculosis, and isolated. I could not stand the idea of bringing stigma to my parents and my family. So, we collectively decided to hide it."¹²¹

"My family still supports me whenever I encounter a problem I cannot solve on my own. I think it is almost impossible for people to overcome such health problems morally and psychologically by themselves."⁶⁸

Tuberculosis is a social disease, so tuberculosis is never just one person's experience. Social stigma can come from health-care providers, employers, community

members, friends, and family, but it can also be experienced by family and household members. When a person has tuberculosis, their loved ones could be vulnerable to infection and stigma. Chronic disease and multimorbidity might be more common in tuberculosis-affected households because of pre-existing risk factors in addition to the direct and indirect effects of tuberculosis. Engaging household and other close contacts for tuberculosis preventive care should be approached with a family-centred integrated approach. Contact investigation is not simply a surveillance strategy with public health interests, but should instead be seen as an opportunity to extend care to a community impacted by tuberculosis and offer therapies such as tuberculosis preventive treatment (TPT) for family members and close contacts with tuberculosis infection.

Considerations for tuberculosis providers

All close contacts of someone with tuberculosis should be counselled and evaluated in a series of care encounters to identify and initiate therapy for people with tuberculosis disease, and to identify people with tuberculosis infection who will benefit from TPT. Losses and dropouts can occur at each step of this cascade of care.¹²² Person-centred care (figure 4) should reduce these losses and enhance the number of people offered TPT.^{123,124}

In the first encounter, contacts should be asked about tuberculosis symptoms and, wherever available, tested for tuberculosis infection.¹²³ Currently, there are two types of tests for tuberculosis infection that measure immune response to *M tuberculosis*: in vivo (cutaneous injection of antigens) and ex vivo (blood-based tests).¹²³ Mathematical modelling suggests most individuals will clear tuberculosis infection within 10 years,¹²⁵ but currently available tests for tuberculosis infection do not distinguish persistence of viable bacilli from cleared infection. Hence, these tests are not useful to verify cure of tuberculosis infection with treatment and should be interpreted considering other epidemiological and clinical information to estimate the risk of disease, and potential benefits of treatment in untreated people. Online decision aids can be useful.^{21,126}

In vivo tests are the TST and the tuberculosis-specific skin tests (TBSTs).¹²⁷ These tests can be done in clinics or at homes; no laboratory infrastructure is needed. In people with previous BCG vaccination, the TST has lower specificity,^{123,128} but this problem is avoided with the recently approved TBST^{123,127,128} as these are based on two antigens, ESAT-6 and CFP-10,¹²⁹ which are more specific to *M tuberculosis*.

Ex vivo tests or IGRAs are laboratory based. Blood is incubated overnight in tubes containing the same ESAT-6 and CFP-10 antigens, along with negative and positive control tubes, and interferon- γ production is measured, either directly in the supernatant using ELISA techniques, or by counting the lymphocytes producing

interferon- γ . At present, only three blood-based assays are approved by WHO for the diagnosis of tuberculosis infection,¹³⁰ although more than a dozen such assays are under development.¹³⁰

There is extensive evidence from trials and observational studies that testing for infection will identify people with positive TST or IGRA who are at highest risk of developing tuberculosis disease,^{21,23,24,35,131} and who will benefit from treatment.^{132–134} From the same studies, there is consistent evidence of lower risk of tuberculosis disease among those who are TST or IGRA negative, and very little evidence of benefit of tuberculosis preventive therapy. A recent individual participant data meta-analysis with a total of 423 497 close contacts of all ages from 25 cohorts revealed that these contacts benefited from TPT when they were TST or IGRA positive (relative risk among treated *vs* untreated 0·17, 95% CI 0·14–0·20), but not if they were TST or IGRA negative (relative risk 0·99, 95% CI 0·67–1·44). However, among the subgroup of children younger than 5 years, TPT was beneficial even if they were TST or IGRA negative.^{134,135}

Tests for tuberculosis infection, including the TST or TBST, require training and quality control, as with any test in clinical medicine, and these tests—as well as tests to exclude disease—can be inaccurate or difficult to interpret, especially in young children. Testing could also create barriers or cause substantial delays in initiating TPT. Recognising these barriers, WHO recommends that testing for tuberculosis before offering TPT is preferred, but is not essential; therefore, in settings where these tests are not available, this should not be a barrier to initiating TPT in individuals at highest risk.¹³⁶ Providers should refer to their country's

recommendations and consider availability of testing when evaluating contacts.

When the tuberculosis infection test result is obtained, usually 48–72 h later, symptoms can be re-evaluated. In people with a positive tuberculosis infection test, before prescribing TPT, a chest radiograph or rapid molecular test should be obtained to rule out tuberculosis disease. Sensitivity of symptom screen is adequate to detect tuberculosis disease in people living with HIV not on antiretroviral therapy,¹³⁷ and in children younger than 5 years,¹³⁸ but sensitivity is only 53% in people living with HIV on antiretroviral therapy⁶⁶ and similarly low in people without HIV.^{12,139} Hence, additional testing is necessary to avoid misdiagnosis and inadvertent monotherapy of tuberculosis. Given the speed at which tuberculosis disease can develop in contacts, especially young children and people living with HIV, it is imperative (and should be possible) that the entire investigation is completed and TPT started within 1–2 weeks.

For decades, the only TPT regimen used was isoniazid for 6–12 months. In multiple randomised trials, this regimen was shown to prevent tuberculosis disease, when given to people with evidence of tuberculosis infection. However, adherence and completion of isoniazid is poor and severe, even fatal, adverse events can occur, particularly hepatotoxicity.

Over the past 20 years, several shorter rifamycin-containing regimens have been recommended by WHO¹³⁶ and other agencies¹⁴⁰ (table 3). All have been shown to have non-inferior efficacy to the longer monoisoniazid regimens, but better completion. Of these regimens, 3 months of isoniazid and rifampicin is attractive as the formulations used for TPT are the same as those used for tuberculosis disease. However, this regimen is not safer

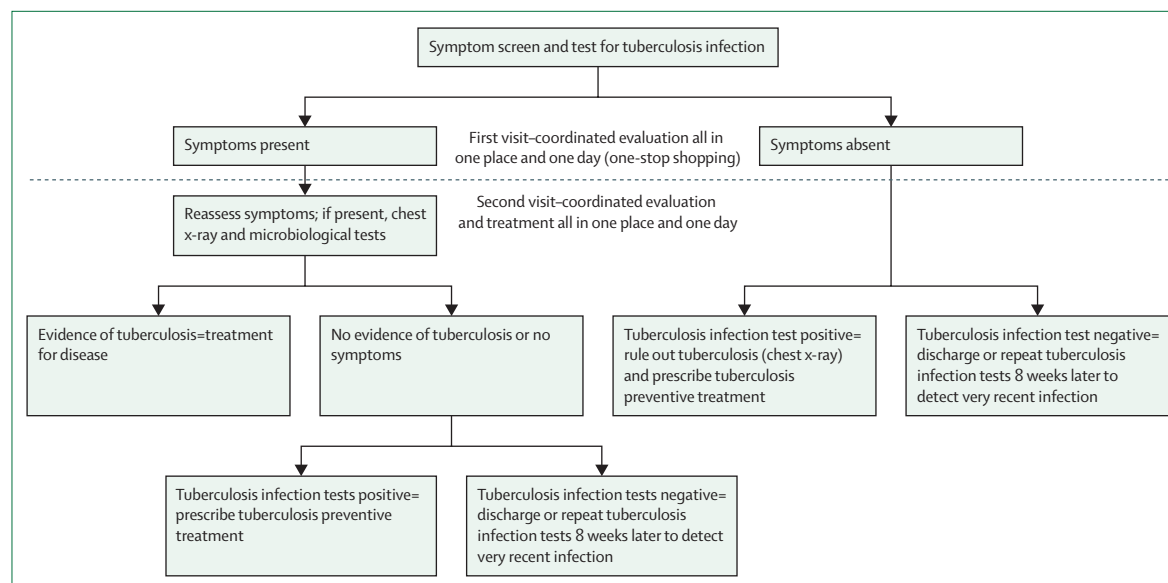


Figure 4: Person-centred integrated algorithm for contact evaluation

than 6 months of isoniazid, and completion has not been consistently better.¹⁴¹ A 12-dose regimen of isoniazid and rifapentine given once per week for 3 months is being scaled up in many countries; this regimen is approved for children aged 2 years and older.¹³⁶ This regimen has excellent efficacy,¹⁴² but is associated with hypersensitivity reactions,¹⁴³ which can be common in some populations,^{144,145} and sometimes severe.^{143,146} 4 months of daily rifampicin has the lowest rates of severe adverse events of all currently recommended TPT regimens^{146,147} and can be used in children of all ages,¹⁴⁸ but completion is lower than with the 3 months weekly isoniazid and rifapentine regimen.¹⁴⁹ A regimen of 1 month isoniazid plus rifapentine given daily had superior completion, and non-inferior efficacy among people living with HIV;¹⁵⁰ evaluation of safety and tolerability in people without HIV infection, children, and other populations is ongoing. For contacts of multidrug-resistant tuberculosis or rifampicin-resistant tuberculosis, a trial in children in South Africa and a trial among adults in Viet Nam, both published in 2024,^{151–154} have concluded that 6 months levofloxacin reduces the risk of subsequent disease by 50–60%.

Antiretroviral therapy for people living with HIV¹⁵⁵ and nutritional support for close contacts in settings with high rates of undernourishment have also been shown to lower the risk of tuberculosis disease.¹⁵⁶ BCG vaccination reduces the risk of infection, progression to disease, and mortality from tuberculosis in children if given in the first month of life, but the effect is insufficient¹⁵⁷ and the duration of protection is still controversial, probably waning with age.^{16,18} Cash transfers can improve TPT completion¹⁵⁸ and should be considered.

Long-term management, follow up, and outcomes

Tuberculosis survivor narrative and perspective

Personal account from We Are Tuberculosis support space meeting (May 15, 2023):

“When I finished treatment, all my pills came to an abrupt stop, and I wondered, ‘what next? How do I take care of myself now?’”

Survivors of tuberculosis, especially after experiencing a physically, socially, and psychologically challenging treatment process, grapple with the difference between being cured of tuberculosis and feeling healed. Post treatment, survivors might wonder, “When will I feel like myself again?”, “Will these side effects go away?”, “How often do I need to get a check-up?”, and “Am I more vulnerable to other illnesses and conditions now that I’ve had tuberculosis?” Providers should consider how tuberculosis and its treatment has created new or exacerbated pre-existing hardships. Insight from survivors could strengthen understanding of the long-term effects, how to minimise them, and improve recovery. More efforts are needed to holistically treat people with tuberculosis with genuine care, dignity, and respect.

Considerations for tuberculosis providers

Typically, tuberculosis disease has been considered a curable illness, with little consideration of the long-term effects for those who have been cured. Hence, for most people treated for tuberculosis, care abruptly ends at treatment completion.¹⁵⁹ This termination of care results in neglect of the long-term clinical, psychological, social, and economic sequelae of tuberculosis on survivors.^{160,161} Given that tuberculosis survivors now number over 155 million globally,¹⁶² the burden of these sequelae is estimated to be similar to the burden associated with the acute phase of tuberculosis disease.¹⁶³

In a meta-analysis, 27083 tuberculosis survivors had a standardised mortality ratio of 3·8 (95% CI 3·0–4·7) after successful tuberculosis treatment, compared with controls, with their most common causes of death being cardiovascular disease and cancer.¹⁶⁴ In a meta-analysis of 41014 primarily adult survivors of pulmonary tuberculosis, 59% had abnormal spirometry compared with 5% of controls.¹⁶⁵ In the subset of survivors with such assessments, nearly 25% reported breathlessness (Medical Research Council breathlessness or dyspnoea scale 3–5) and 6-min walk distance was 79% of predicted, suggesting potential effects on functional ability.¹⁶⁵ As many as two-thirds of children and adolescents surviving pulmonary tuberculosis could develop respiratory impairment and children developing tuberculosis in their first 5 years of life are about twice as likely to have difficulties with wheezing and coughing when compared with those without tuberculosis.^{166,167} Tuberculosis survivors are also substantially more likely to develop lung cancer, chronic obstructive pulmonary disease, aspergillomas, and bronchiectasis.^{161,165,168}

Tuberculous meningitis and CNS tuberculosis might result in lasting neurological abnormalities including cognitive impairment, cerebral palsy, seizures, or blindness, in up to half of children and adolescents,¹⁶⁶ and one-quarter to one-third of adults.¹⁶⁹ Musculoskeletal tuberculosis in adults and children might lead to lasting sequelae, such as bone deformities and chronic pain, and tuberculosis affecting the spine can lead to kyphosis and even paraplegia.^{166,170}

People diagnosed with tuberculosis also face numerous social stressors, such as stigma, loss of income, isolation in the early stages of treatment, and absence of social support. In a prospective study in Malawi, schoolchildren had frequent interruptions in schooling during treatment and for 1 year thereafter, while adult survivors earned and worked less after than before their tuberculosis diagnosis.¹⁷¹ Approximately 20% of tuberculosis survivors also report mental health disorders that persist after treatment.¹⁷²

Recent clinical practice guidelines for post-tuberculosis lung disease¹⁰⁸ recommend assessing every person completing tuberculosis treatment for post-tuberculosis lung disease as soon as possible after treatment ends. This evaluation could include some of the following

assessments: clinical status (symptoms); health status (quality of life, Patient Health Questionnaire [PHQ-9]); exercise capacity (Medical Research Council dyspnoea scale, 6-min walk test); lung imaging (chest radiograph); and lung function (spirometry).¹⁰⁸ Recommendations for pulmonary rehabilitation suggest treatment should include aerobic activities, use of airway clearing techniques, and nutritional support, for which there is good evidence of benefit,^{108,173} and also provide comprehensive health education programmes covering smoking cessation and other topics.¹⁷⁴ Other approaches, such as inspiratory muscle training, could be considered.¹⁷⁴

Controversies, uncertainties, and outstanding research questions

As with many areas of clinical medicine, in tuberculosis the greatest areas of uncertainty are also the most controversial; fortunately, they are also the most active areas of research. Diagnosis of tuberculosis disease is slow, expensive, and inaccessible in many settings. Ongoing research is testing alternative samples such as saliva, breath-based and swab-based specimens,¹⁷⁵ and use of artificial intelligence algorithms to interpret recorded cough sounds.¹⁷⁶ Whole-genome sequencing detects *M tuberculosis*, plus mutations responsible for resistance,¹⁷⁷ but is costly and requires complex infrastructure, hampering implementation where it is needed most.¹⁷⁸ Targeted next-generation sequencing, is promising as it is performed directly on clinical samples, and amplifies specific DNA sequences to detect tuberculosis, and resistance at lower cost.¹⁷⁹

Treatment of drug-susceptible tuberculosis disease remains stuck at 6 months for most patients, despite many treatment-shortening trials, which have involved giving higher doses of anti-tuberculosis drugs, or a greater number of these drugs, including new anti-tuberculosis drugs. To date, these trials have shown a trade-off between shorter versus safer or better tolerated.⁷⁹ New drugs that permit treatment shortening and are also safe and well tolerated are needed; many are in the pipeline.¹⁸⁰ Treatment of drug-resistant tuberculosis has leapt forward in the past decade with the development of all-oral, 6-month regimens that are effective and safer than previous regimens. Further treatment shortening will also depend upon new drugs.

Tuberculosis prevention remains a major challenge, given the absence of an effective vaccine, poor diagnostics, and lengthy plus potentially toxic preventive treatment. A phase 3 study¹⁸¹ has been initiated to test a new vaccine that had 50% efficacy for prevention of tuberculosis disease in IGRA-positive adults in a phase 2 trial.¹⁸² Candidate vaccines for preventing tuberculosis infection, disease, or recurrence are being tested,¹⁸³ or are in the pipeline.^{184,185} Current tests for tuberculosis infection are unsatisfactory, as very few with a positive test (TST or IGRA) will ever develop disease,

leading to controversy as to whether these tests are needed. However imperfect, currently available tests do identify people at higher risk for tuberculosis disease, and also those who will benefit from preventive therapy. Improved tests that would identify people with highest risk for tuberculosis disease, by improving identification of people who will derive maximal benefit from preventive therapy, would be a major advance. Development of such tests is still in the early stages. In the absence of better tests, the pressing need is to develop and evaluate safer and better tolerated tuberculosis preventive regimens of 1–2 months duration.

The need for continued patient care for tuberculosis-related disability after successful treatment of tuberculosis disease has only recently been recognised. Better understanding of who is at risk for these sequelae, and why, plus how to prevent or mitigate them, possibly by modulating host response, will inform future therapeutic trials.

Conclusion

Tuberculosis continues to be a major challenge for all who confront it. For people with tuberculosis, challenges include being diagnosed quickly to reduce disease progression, and being treated with an effective, safe, and well tolerated treatment that is easy to complete. For care providers, challenges include thinking about tuberculosis, and initiating testing that is rapid, yet accurate, then selecting the shortest, safest, and most effective treatment. A key provider role is to educate people with tuberculosis and their families to reduce shame, stigma, confusion, and fear, and to organise adequate treatment support to ensure full adherence with the selected regimen. The challenges for tuberculosis programmes include prevention of disease through strategies to address social determinants, and detection and treatment of tuberculosis infection, and to reduce transmission, morbidity, and mortality through improved detection and treatment of tuberculosis disease. For researchers, development and application of new tools for diagnosis, treatment, and prevention, based on improved understanding of the biological mechanisms of infection and disease, pose the biggest challenges. Progress over the past decade leads us to be optimistic that these challenges can be overcome, and tuberculosis ultimately eliminated.

Contributors

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Declaration of interests

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Figures 1 and 2 adapted from Furin et al.¹⁸⁶ Figure 3 created with Canva. Table 1 adapted from Campbell et al.,¹⁸⁷ by permission of the authors.

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